

Team ML-ATE: Final Charter

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1. Project Title and Overview

Project Title: Project LiAISon

Project LiAISon focuses on detecting stationary anomalies in fused sonar–LiDAR point clouds to support maritime safety and underwater infrastructure monitoring. All point clouds used in this project are synthetically generated through open-source 3D modeling tools (e.g., Blender or BIAnder for sensor simulation), allowing controlled simulation of underwater environments and irregular structures such as debris, deformities, or unnatural formations.

A separate team is responsible for producing the fused sonar–LiDAR data. Our scope begins once the fused point clouds are provided, enabling us to focus on developing preprocessing workflows, implementing anomaly detection methods, and evaluating detection results using synthetic ground-truth scenarios. By narrowing the scope to detection rather than data fusion, the project aims to prototype techniques that can inform future automated inspection systems and reduce reliance on manual diver-based assessments.

2. Problem Statement

Underwater inspections of piers, ports, and coastal infrastructure currently rely heavily on manual diver-based methods. These inspections are time-consuming, hazardous, and often yield inconsistent or incomplete data due to environmental limitations such as poor visibility, currents, and equipment constraints. As a result, organizations face increased operational risk, delayed maintenance decisions, and higher costs for repairs.

By developing an anomaly detection pipeline, this project seeks to improve inspection speed, accuracy, and operational safety.

3. Project Goals and Objectives

Project Goal: Develop an anomaly detection workflow for fused sonar–LiDAR point clouds generated using open-source 3D modeling tools.

Objectives:

- **Synthetic Dataset Creation**
 - Generate synthetic underwater environments in Blender or BIAnder and export fused point clouds in standard formats such as .ply or .pcd.
 - Include embedded ground-truth anomalies (e.g., debris, structural irregularities) to support evaluation.
- **Preprocessing Pipeline**
 - Clean and normalize fused point clouds using noise filtering, downsampling, and alignment.
 - Standardize data formats and metadata to ensure the workflow is reproducible.
- **Anomaly Definition and Labeling**
 - Define anomaly categories relevant to underwater infrastructure
 - Create ground-truth labels for the synthetic datasets.
- **Detection Algorithm Development**
 - Implement and compare multiple anomaly-detection approaches, such as machine learning models, statistical outlier detection, and geometric feature-based methods.
- **Visualization Tools**
 - Develop 3D visualization outputs that highlight detected anomalies through color coding or overlays.
- **Evaluation and Validation**
 - Evaluate detection performance using precision, recall, F1 score, false-positive and false-negative rates, and runtime efficiency.
 - Validate results against the ground-truth anomalies embedded in the synthetic datasets.

4. Stakeholders and Roles

Sponsor - MerLion Advisory: Provides project direction.

Team - ML-ATE: Responsible for developing and implementing the project solution.

- Leisha Khapre - Leader / Quality Assurance / Reviewer
- Maddie Bird - Project Management / Analyst / Documenter
- Inayaa Khoja - Project Management / Analyst / Documenter
- Jade Winebright - Documenter / Reviewer / Developer
- Gaurav Shatoor - Developer / Designer / Researcher

Team Communication: Weekly meetings, shared Google Drive; will use GitHub for version control.

Instructor - Dr. Abuabara: Guides milestones and deliverables.

Advisors - Dr. Garcia, Anshul Yadav: Provide technical guidance, review deliverables, and offer feedback.

5. Timeline and Major Milestones

- **Aug – Nov 2025:**
Initial project exploration and development of charter drafts.
- **Dec 2025:**
Finalize and present the completed project charter.
- **Dec 2025 – Jan 2026:**
Create synthetic underwater scenes and generate sonar–LiDAR datasets. Begin preprocessing and standardizing fused point cloud formats.
- **Feb – Mar 2026:**
Define anomaly categories and develop prototype detection algorithms. Conduct initial tests and refine

preprocessing workflows as needed.

- **Mar – Apr 2026:**

Evaluate detection performance using ground-truth anomalies and defined metrics. Refine algorithms based on results. Potentially produce 3D visualizations of detected anomalies and complete system integration.

- **May 2026:**

Finalize documentation, validation summary, and deliver the final report and sponsor presentation.

6. Deliverables

- Synthetic sonar–LiDAR datasets generated from 3D underwater scenes, including raw and fused point clouds in standard formats (e.g., .ply, .pcd).
- Preprocessed and labeled point cloud datasets with documented anomaly ground truth.
- Anomaly detection scripts or notebooks implementing all tested algorithms, along with a reproducible code repository and environment files.
- 3D visualizations of raw, preprocessed, and anomaly-highlighted point clouds.
- Validation report summarizing performance metrics (precision, recall, F1 score, false-positive rate, runtime).
- Final documentation describing system design, methods, dataset-generation process, assumptions, and future improvements.

7. Success Criteria and Metrics

- Data Preparation
 - Point cloud data cleaned, organized, and prepared for analysis.
- Anomaly Definitions
 - Clear categories of stationary anomalies with corresponding ground-truth labels in synthetic data.
- Workflow Functionality
 - A complete pipeline that runs from input point clouds to anomaly outputs.
- Detection Performance
 - Algorithms perform above baseline methods using metrics such as precision, recall, F1 score, and false-positive rate (no fixed thresholds defined yet).
- Visualization
 - Visual tools clearly display detected anomalies.
- Documentation & Reproducibility
 - The entire workflow is fully documented and reproducible.

8. Assumptions and Risks

Assumptions

- Software used for synthetic data provides sufficient fully fused point clouds.
- Resolution is sufficient for anomaly detection.
- The workflow will still be valuable to MerLion although the project uses synthetic data.

Risks

Risk	Likelihood	Category	Owner	Mitigation
File loss	2	Medium	All Team Members	Using git commits often and backing up code
Scheduling Risks	3	High	Project Managers	Maintaining and updating timeline, conducting phase reviews

Synthetic ground truth may not match real conditions.	4	Medium	Developers	Document uncertainty levels, and design synthetic scenarios that mimic real-world conditions.
Misinterpretation of outputs	3	Low	Project Managers	Using non technical summaries, regular meetings with stakeholders
Computational load	3	Medium	Team Members	Mitigate via downsampling or cloud computing

9. Ethical, Legal Standards and Constraints

The team will follow university standards and handle all project data securely and responsibly. All simulated datasets used for testing will be clearly labeled. All work will adhere to the principles of transparency, accountability, and reproducibility.

Sponsor Signature: _____

Team Initials: MB, LK, JW, GS, IK